Probability and Statistics

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Exercise: Given a uniform distribution $U(a,b)$ where a is the center and b is the total width, how many data points does one need to have a 95% confidence interval for the average have a radius of less than 5% of the width?
Let S be a sample of n datapoints sampled randomly from $U(a,b)$. For two points, what is the distribution for the sum of two? A single distribution has the formula:

$$f_U(x) = \begin{cases} 0, & x < 0 \text{ or } x > 1 \\ 1, & 0 \leq x \leq 1 \end{cases}$$

Recall the following:

Theorem 1. The pdf for the distribution of $X + Y$ of two random variables is given by

$$f_{X+Y}(x) = \int_{-\infty}^{\infty} f_X(t) f_y(x-t) dt = f_X * f_Y$$

Thus we get

$$f_{2U}(x) = \int_{-\infty}^{\infty} 1_{[0,1]}(t) 1_{[0,1]}(x-t) dt$$

Now note that

$$1_{[0,1]}(t)1_{[0,1]}(x-t) = \begin{cases} 0 & \text{if } t \notin [0,1] \text{ or } x-t \notin [0,1] \\ 1 & \text{if } t, x-t \in [0,1] \end{cases} = 1_{[x-1,x]\cap[0,1]}$$